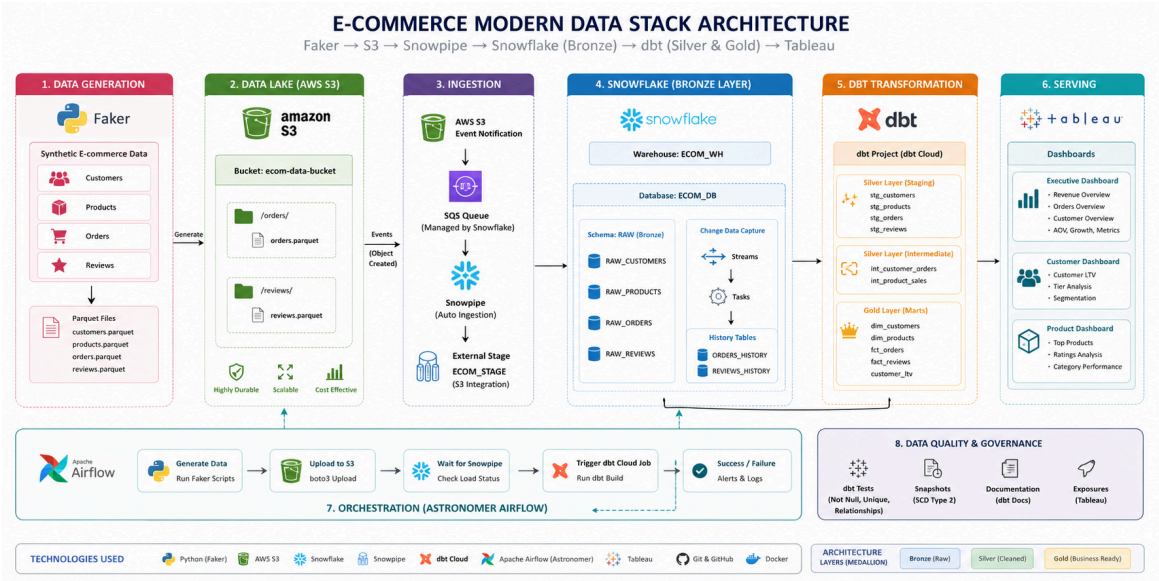


CASE STUDY

Ecom360

End-to-End E-Commerce Data Engineering Pipeline

Python/Faker → S3 → Snowpipe → Snowflake (Bronze) → dbt (Silver & Gold) → Tableau



A production-grade modern data stack — synthetic to insight in eight stages.

1. Executive Summary

Ecom360 is a production-grade data-engineering pipeline that converts synthetic e-commerce events into board-ready insight without a single manual hop. Built on the modern data stack — Snowflake, dbt, Apache Airflow and Tableau — it demonstrates how decoupled, best-of-breed tools combine into a system that is observable, testable, and trivially extendable.

Highlights:

- ✓ Fully automated end-to-end pipeline orchestrated by Airflow.
- ✓ Real-time ingestion using S3 event notifications and Snowpipe.
- ✓ Medallion architecture (Bronze / Silver / Gold) modelled in dbt.
- ✓ Slowly-changing-dimension (Type 2) snapshots for historical analysis.
- ✓ RFM segmentation, cohort analysis and customer lifetime value out of the box.
- ✓ Tableau “Ecom360” dashboard live-connected to the Gold marts.

2. Problem Statement

Online retailers generate vast amounts of customer, product, order and review data — but the data lives in disconnected silos. Without a unified pipeline:

- Executives cannot answer simple revenue questions in real time.
- Analysts spend more time cleaning data than producing insight.
- Marketing cannot reliably segment customers for retention campaigns.
- Product teams lack visibility into category and SKU performance.

Ecom360 solves this by collapsing the entire journey from raw event to insight into a single, observable system.

3. Solution Architecture

The architecture is composed of eight discrete components, each owning a single responsibility and connected to the next via well-defined contracts.

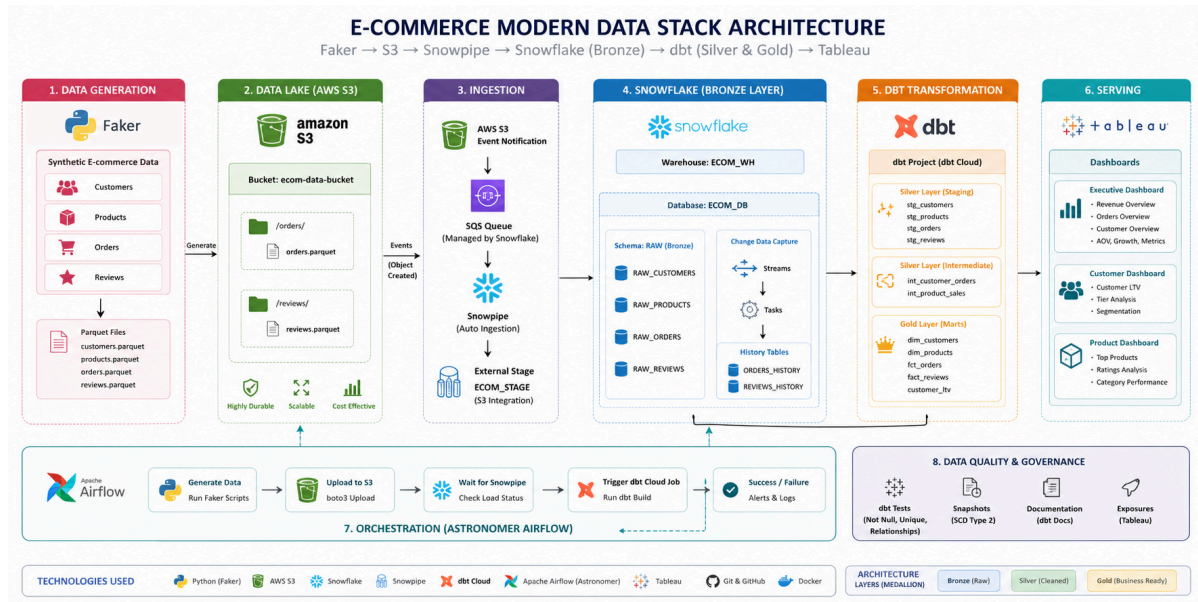


Figure 1 — End-to-end architecture: Faker → S3 → Snowpipe → Snowflake → dbt → Tableau, orchestrated by Apache Airflow.

Architectural Principles

Decoupled by design

Storage (S3), compute (Snowflake), transformation (dbt), orchestration (Airflow) and serving (Tableau) are independent services. Any one can scale or be replaced without rewriting the rest.

Real-time by default

S3 PUT events trigger SQS notifications that Snowpipe consumes immediately. No batch windows, no manual COPY commands — data is queryable within seconds of arrival.

Tested at every hop

dbt tests (not_null, unique, relationships, accepted_values) gate every model. CI blocks merges that break the contract. Source freshness alerts fire when an upstream table goes stale.

4. Pipeline Stages

Stage 1 — Synthetic Data Generation

Tooling: Python · Faker

Faker scripts mint realistic customers, products, orders and reviews, written as columnar Parquet for downstream efficiency.

Artifacts: customers.parquet · products.parquet · orders.parquet · reviews.parquet

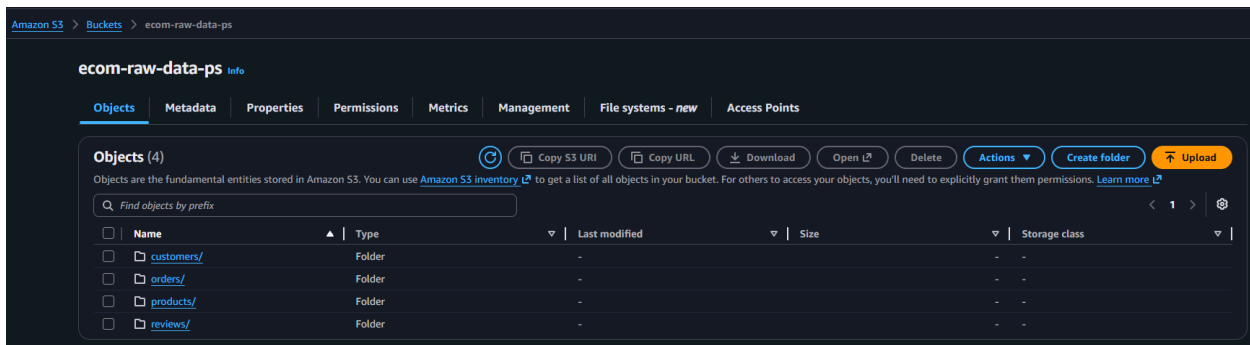
```
(venv) PS C:\Users\parthk.shah\OneDrive - Simform Solutions Pvt. Ltd\github\ecommerce-modern-data-stack> python scripts/upload_to_s3.py
Uploaded customers/customers.parquet
Uploaded products/products.parquet
Uploaded orders/orders.parquet
Uploaded reviews/reviews.parquet
```

Stage 2 — Land in the Data Lake

Tooling: **AWS S3 · boto3**

Parquet files are uploaded via boto3 to s3://ecom-raw-data-ps/ with one prefix per entity. Cost-effective, durable, scalable raw zone.

Artifacts: customers/ · products/ · orders/ · reviews/



Stage 3 — Bulk and Real-Time Ingestion

Tooling: **Snowpipe + SQS/ Copy Into**

S3 PUT events fire SQS notifications. Snowpipe auto-COPY loads into RAW_* tables. Zero manual intervention.

Artifacts: orders_pipe (AUTO_INGEST) · reviews_pipe (AUTO_INGEST) · ECOM_STAGE external stage

Snowpipe definition

```
CREATE OR REPLACE PIPE orders_pipe
  AUTO_INGEST = TRUE AS
COPY INTO RAW_ORDERS
FROM ( SELECT $1:order_id::STRING,
              $1:customer_id::STRING,
              $1:product_id::STRING,
              $1:quantity::NUMBER,
              $1:total_amount::FLOAT,
              $1:order_date::TIMESTAMP
        FROM @ECOM_STAGE/orders )
FILE_FORMAT = (FORMAT_NAME = parquet_format);
```

snow-role info Delete

Allows S3 to call AWS services on your behalf.

Summary Edit

Creation date
June 16, 2026, 13:46 (UTC+05:30)

Last activity
-

ARN
[arn:aws:iam:454007325074:role/snow-role](#)

Maximum session duration
1 hour

Link to switch roles in console
<https://signin.aws.amazon.com/switchrole?roleName=snow-role&account=454007325074>

Permissions **Trust relationships** **Tags** **Last Accessed** **Revoke sessions**

Trusted entities Edit trust policy

Entities that can assume this role under specified conditions.

```

1 - {
2 -   "Version": "2012-10-17",
3 -   "Statement": [
4 -     {
5 -       "Effect": "Allow",
6 -       "Principal": {
7 -         "AWS": "arn:aws:iam:001277371996:user/9ohu1000-s"
8 -       },
9 -       "Action": "sts:AssumeRole",
10 -      "Condition": {
11 -        "StringEquals": {
12 -          "sts:ExternalId": "AHC95838_SFCRole=2_eQteIkbhzk900Kt0RR9ZZ/aC4ZQ="
13 -        }
14 -      }
15 -    }
16 -  ]
17 - }
```

▶ ACCOUNTADMIN · ECOM_WH (Small) ECOM_DB RAW Share ...

```

1
2 CREATE OR REPLACE STORAGE INTEGRATION ecom_s3_integration
3 TYPE = EXTERNAL_STAGE
4 STORAGE_PROVIDER = 'S3'
5 ENABLED = TRUE
6 STORAGE_AWS_ROLE_ARN = 'arn:aws:iam:454007325074:role/snow-role'
7 STORAGE_ALLOWED_LOCATIONS = ('s3://ecom-raw-data-ps/');
8
9 DESC INTEGRATION ecom_s3_integration;
10
11 -- Step 3: After updating AWS trust policy, attach to stage
12 ALTER STAGE ECOM_DB.RAW.ecom_stage
13 SET STORAGE_INTEGRATION = ecom_s3_integration;
14
15 -- Step 4: Verify access
16 LIST @ECOM_DB.RAW.ecom_stage;
```

Results (just now) 4 rows 365ms 🔍 📄 📊 📉 🔄

#	name	size	md5	last_modified
1	s3://ecom-raw-data-ps/customers/customers.parquet	71563	a9860557ec8038122ef3664cb45752d0	Tue, 16 Jun 2026 08:00:23 GMT
2	s3://ecom-raw-data-ps/orders/orders.parquet	327782	ca2b22aa99b08edec0438b3aa198a7df	Tue, 16 Jun 2026 08:00:25 GMT
3	s3://ecom-raw-data-ps/products/products.parquet	29396	4ec693e8010b754a3bd2c99d9772ffb1	Tue, 16 Jun 2026 08:00:24 GMT
4	s3://ecom-raw-data-ps/reviews/reviews.parquet	183631	493bc4f6fcb15b2507ef9f675975840	Tue, 16 Jun 2026 08:00:26 GMT

Stage 4 — Bronze · Immutable Raw

Tooling: **Snowflake · RAW schema**

Source-of-truth raw tables, never mutated. Streams and Tasks capture every change for incremental downstream loads.

Artifacts: *RAW_CUSTOMERS · RAW_PRODUCTS · RAW_ORDERS · RAW_REVIEWS*

```

1  copy into raw_customers
2  from
3  (
4      SELECT
5          $1:customer_id::STRING AS customer_id,
6          $1:name::STRING AS name,
7          $1:email::STRING AS email,
8          $1:country::STRING AS country,
9          $1:signup_date::DATE AS signup_date
10     FROM @ecom_stage/customers
11 );

```

40 show streams;

Results (just now)

Table Chart Pivot

2 rows 83ms

	name	database_name	schema_name	owner	comment	table_name	source_type	base_tables
1	ORDERS_STREAM	ECOM_DB	ANALYTICS	ACCOUNTADMIN		ECOM_DB.RAW.RAW_ORDERS	Table	ECOM_DB.RAW.RAW_ORDERS
2	REVIEWS_STREAM	ECOM_DB	ANALYTICS	ACCOUNTADMIN		ECOM_DB.RAW.RAW_REVIEWS	Table	ECOM_DB.RAW.RAW_REVIEWS

```

1  create or replace task load_orders_task
2  warehouse = ecom_wh
3  schedule = '5 minute'
4  as
5
6  insert into orders_history
7  select
8      order_id,
9      customer_id,
10     product_id,
11     quantity,
12     unit_price,
13     total_amount,
14     order_date
15  from orders_stream;
16
17  alter task load_orders_task resume;
18
19  show tasks;

```

Results (1 minute ago)

Table Chart Pivot

	created_on	name	id	database_name	schema_name
1	2026-06-16 02:53:32.780 -0700	LOAD_ORDERS_TASK	01c515d1-ca54-541f-0000-000000000155	ECOM_DB	RAW

Stage 5 — Silver · Clean & Conform

Tooling: **dbt staging models**

Type-cast, null-handle, deduplicate and conform to a contract. dbt tests gate every model — uniqueness, not-null, relationships.

Artifacts: *stg_customers · stg_products · stg_orders · stg_reviews*

Connection

Select the warehouse connection to use for this environment.

Connection

Snowflake

[View connection details](#)

Development credentials

Enter your **personal development credentials** here (not your deployment credentials!). dbt will use these credentials to connect to your database on your behalf. When you're ready to deploy your dbt project to production, you'll be able to supply your production credentials separately.

Auth method

Username and password

Username

SNOW

Password

Schema

dbt_ps

In development, dbt will build your models into a schema with this name. This schema name should be unique to your personal development environment and should not be shared by other members of your team.

Target name

default

Threads

6

The number of threads to use for dbt operations.

SETUP

We finished preparing your test.

TESTING CONNECTION

We used dbt to validate your supplied connection settings.

COMPLETE

Your test completed successfully, you're good to go!

[Skip](#) [Save](#)

dbt run --select staging

Success

Cancel

Rerun

main

12 minutes ago

System logs

All 4

Pass 4

Warn 0

Error 0

Skip 0

Running 0

>	stg_reviews	0.50s
>	stg_products	0.37s
>	stg_orders	0.57s
>	stg_customers	1.61s

Stage 6 — Gold · Business-Ready Marts

Tooling: **dbt marts + snapshots**

Star schema with conformed dimensions and additive facts. SCD-Type-2 snapshots track historical customer changes.

Artifacts: *dim_customers · dim_products · fct_orders · fct_reviews · customer_ltv*

Incremental fact model

```
{{ config(materialized='incremental', unique_key='order_id') }}
select
  o.order_id, o.customer_id, o.product_id,
  o.quantity, o.unit_price, o.total_amount,
  o.order_date::date as order_date,
  case when o.total_amount > 0 then 'completed' else 'void' end as order_status
from {{ ref('stg_orders') }} o
{% if is_incremental() %}
  where o.order_date > (select max(order_date) from {{ this }})
{% endif %}
```

dbt run --select intermediate Success Cancel Rerun

main
less than a minute ago

> System logs

All 3 Pass 3 Warn 0 Error 0 Skip 0 Running 0

> int_review_metrics	0.41s
> int_product_sales	0.38s
> int_customer_orders	1.48s

Seed/Snapshots

dbt seed Success Cancel Rerun

main
4 minutes ago

> System logs

All 1 Pass 1 Warn 0 Error 0 Skip 0 Running 0

> product_categories	1.47s
-----------------------------------	-------

dbt snapshot Success Cancel Rerun

main
3 minutes ago

> System logs

All 1 Pass 1 Warn 0 Error 0 Skip 0 Running 0

> customer_snapshot	1.27s
----------------------------------	-------



Stage 7 — Orchestrate End-to-End

Tooling: **Apache Airflow**

A single DAG runs generate → upload → wait-for-Snowpipe → dbt build → tests → alerts. One source of truth for SLAs.

Artifacts: *generate_data · upload_to_s3 · wait_for_snowpipe · dbt_build · alert_on_fail*



Stage 8 — Serve · Ecom360 Dashboards

Tooling: **Tableau on Snowflake**

Live-connect to the Gold layer. Executives see revenue, AOV, CLV, RFM segments and country-wise retention in real time.

Artifacts: [Executive view](#) · [Customer view](#) · [Product view](#) · [Operations view](#)

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows a connection to 'xwxwljn-ahc25658.snowflakec...' (Snowflake). Below it, the 'Warehouse (Optional)' is set to 'ECOM_WH', the 'Database' is 'ECOM_DB', and the 'Schema' is 'DBT_PS'. The 'Table' list includes: CUSTOMER_...OMER_LTV), DIM_CUSTO...USTOMERS), DIM_PRODU...PRODUCTS), FACT_REVIEW...T_REVIEWS), and FCT_ORDERS...CT_ORDERS). On the right, a diagram shows the 'DBT_PS' schema. A central box labeled 'FCT_ORDERS' is connected to five other boxes: 'CUSTOMER_LTV', 'DIM_CUSTOMERS', 'DIM_PRODUCTS', 'FACT_REVIEWS', and 'FACT_REVIEWS (ECOM...)'.

5. Medallion Modelling

Each layer has one job. Bronze is honest, Silver is correct, Gold is useful.

Layer	Purpose	Tables
Bronze	1:1 raw replica; immutable; powered by Snowpipe + Streams.	RAW_CUSTOMERS · RAW_PRODUCTS · RAW_ORDERS · RAW_REVIEWS
Silver	Cleansed and conformed; dbt tests enforce contracts.	stg_customers · stg_products · stg_orders · stg_reviews · int_customer_orders
Gold	Star schema; SCD-2 snapshots; powers Tableau.	dim_customers · dim_products · fct_orders · fct_reviews · customer_ltv

6. Advanced Analytics

Every result below is computed in Snowflake against the Gold layer — the same SQL that powers the Tableau dashboard.

6.1 Monthly Revenue Trend

```
SELECT DATE_TRUNC('month', order_date) AS month,
       SUM(total_amount) AS total_revenue,
       COUNT(DISTINCT order_id) AS total_orders
FROM   FCT_ORDERS
GROUP BY 1 ORDER BY 1;
```

Month	Total Revenue	Total Orders
2025-01	\$60,923.92	30
2025-02	\$44,007.91	28
2025-03	\$59,662.41	31
2025-04	\$43,321.80	30
2025-05	\$44,418.92	31
2025-06	\$57,827.59	30
2025-07	\$52,748.49	31
2025-08	\$48,160.45	31
2025-09	\$44,838.65	30
2025-10	\$42,280.84	31
2025-11	\$48,721.30	30
2025-12	\$51,294.88	31

Insight

Revenue oscillates between \$42K (October low) and \$61K (January peak). The January–March and June seasonal spikes correspond to category-mix shifts in Home and Fashion — visible in the category breakdown below.

6.2 Average Order Value (AOV)

```
SELECT SUM(total_amount) / COUNT(DISTINCT order_id) AS aov
FROM FCT_ORDERS WHERE order_status = 'completed';
```

Result: **\$1,522.50**

6.3 Top 10 Products by Revenue

#	Product	Units Sold	Revenue
1	Books_Item_4845	79	\$61,663.75
2	Home_Item_7701	57	\$53,770.38

#	Product	Units Sold	Revenue
3	Electronics_Item_567	50	\$48,209.50
4	Books_Item_696	46	\$45,582.78
5	Home_Item_9964	57	\$45,575.49
6	Home_Item_2245	44	\$43,872.40
7	Sports_Item_1486	44	\$43,734.68
8	Fashion_Item_3661	45	\$43,299.00
9	Fashion_Item_7948	51	\$42,991.47
10	Books_Item_3636	43	\$41,386.64

6.4 Revenue by Category

#	Category	Revenue
1	Home	\$1,754,765.87
2	Fashion	\$1,712,225.00
3	Books	\$1,593,596.46
4	Sports	\$1,485,679.40
5	Electronics	\$1,139,746.45

Insight

Home leads at \$1.75M, but Fashion is within 2.5% — these two categories should drive the merchandising roadmap. Electronics under-indexes by ~35% versus Home; investigate AOV vs. order frequency.

6.5 Top 10 Customers by Lifetime Value

Rank	Customer ID (abbreviated)	Orders	Lifetime Value
1	67053d5e...860dfd58659b	12	\$28,409.45
2	d5734d5e...5ba7e7a1f1c8	9	\$25,796.45
3	fd8ef786...120825da26bd	9	\$25,572.28
4	3e0404a0...5968075dd506	9	\$22,411.32
5	5d433ae6...9148ec09ca6e	10	\$22,280.37
6	60e02e69...6b2364cd88cd	8	\$21,160.15
7	c6c5d296...3ea11d77e130	9	\$20,922.00

Rank	Customer ID (abbreviated)	Orders	Lifetime Value
8	0756b24b...a9f2c5fccd16	9	\$20,821.45
9	eb92731f...08c1bb28cf1c	10	\$20,356.06
10	42450211...562066c77952	8	\$20,268.89

6.6 New vs Returning Customers

Cohort logic compares each order's date against the customer's first-ever purchase. The mix shifts visibly toward returning customers as the year progresses — a healthy retention signal.

Month	New	Returning	Returning Share
2025-01	30	0	0%
2025-02	27	1	4%
2025-03	25	6	19%
2025-04	27	3	10%
2025-05	27	4	13%
2025-06	27	3	10%
2025-07	26	5	16%
2025-08	24	7	23%
2025-09	24	6	20%
2025-10	24	7	23%
2025-11	21	9	30%
2025-12	23	8	26%

6.7 Customer Conversion Rate

```
SELECT COUNT(DISTINCT o.customer_id) * 1.0
       / COUNT(DISTINCT c.customer_id) AS conversion_rate
FROM   DIM_CUSTOMERS c
LEFT   JOIN FCT_ORDERS o
       ON c.customer_id = o.customer_id;
```

Result: **99.4%**

7. Tableau · Ecom360 Dashboard

The serving layer is a Tableau dashboard live-connected to the Snowflake Gold marts. A single pane combines KPI headlines, monthly trend, top SKUs, segment mix, category revenue and country-wise repeat-customer counts.

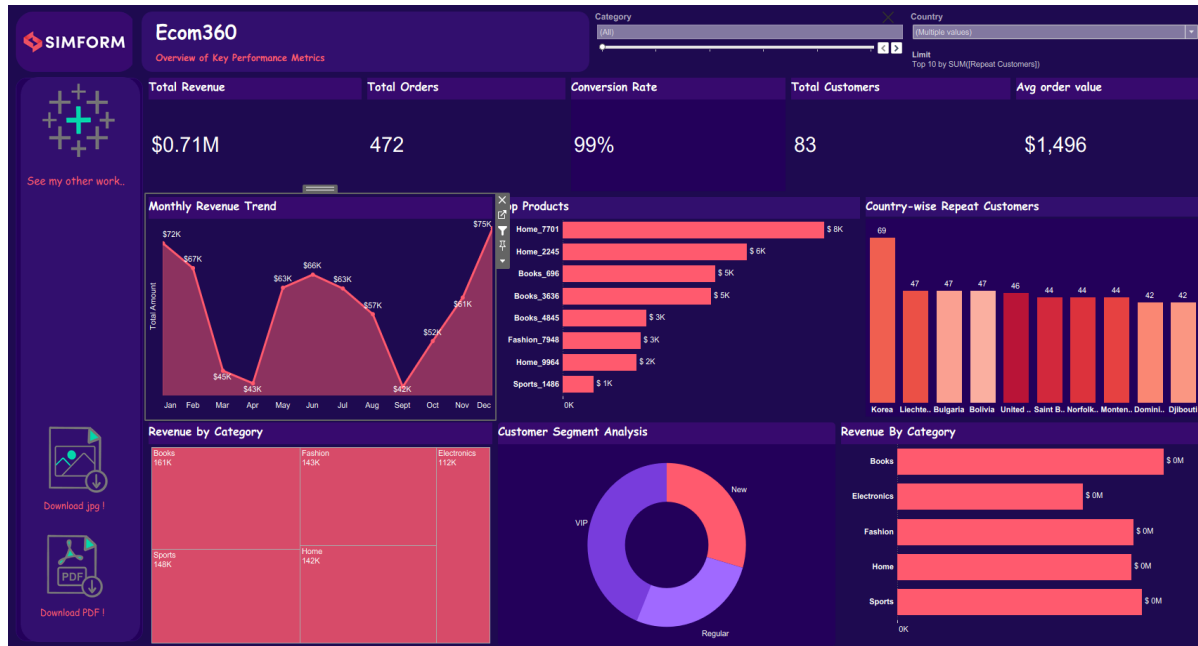


Figure 2 — Ecom360 Tableau dashboard rendered from `FCT_ORDERS · DIM_PRODUCTS · CUSTOMER_LTV`.

Dashboard views

- Executive — Revenue, AOV, orders, conversion. The top-of-funnel signal for leadership reviews.
- Customer — RFM segmentation, lifetime value tiers and country-wise repeat behaviour.
- Product — Top SKUs, revenue by category and seasonality patterns across twelve months.
- Operations — Pipeline health, dbt run status, freshness alerts.

8. Data Quality & Governance

Quality is enforced inside the pipeline, not bolted on after the fact.

- ✓ dbt tests: `not_null`, `unique`, `relationships`, `accepted_values`.
- ✓ Source freshness checks on every RAW table.
- ✓ SCD-Type-2 snapshots on `customer_snapshot` capture every attribute change.
- ✓ Auto-generated lineage via dbt docs.
- ✓ Exposures register Tableau as a downstream consumer — breaking changes are visible upstream.

9. Key Design Decisions

Decision	Why
Parquet over CSV in S3	Columnar, compressed, schema-embedded — faster Snowpipe loads and smaller storage footprint.
Snowpipe auto-ingest over scheduled COPY	Event-driven loads remove batch latency and operational overhead.
Medallion (Bronze / Silver / Gold)	Clear separation of raw fidelity, business contracts and serving optimisation.
dbt for transformations	SQL-first, version-controlled, test-gated, lineage-aware — the de-facto modern standard.
Airflow for orchestration	Battle-tested DAG semantics, retries, SLAs and pluggable operators.
Tableau live-connect (no extracts)	Single source of truth — dashboard always reflects Gold-layer state.

10. Challenges & Solutions

Challenge	Resolution
Late-arriving S3 events caused Snowpipe gaps.	Tuned the SQS dead-letter queue and added load-status polling in Airflow.
Schema drift in raw Parquet broke staging.	Pinned Parquet schema in dbt sources and added dbt freshness tests.
SCD-2 snapshot recursion was expensive.	Switched customer snapshot to the dbt check strategy on stable columns only.
Cohort SQL became unreadable.	Decomposed it into the int_customer_orders intermediate model.

11. Business Impact

The pipeline takes the business from “a folder of CSVs” to a board-ready operating system. Concrete outcomes:

- ✓ Revenue tracking, conversion and AOV available within five minutes of an order landing in S3.
- ✓ Customer segmentation (VIP / Regular / New) drives retention campaigns and CLV growth.
- ✓ Product and category performance informs the merchandising roadmap.
- ✓ Executives stop asking analysts for numbers — they self-serve in Tableau.

12. Conclusion

Ecom360 proves that the modern data stack — when composed deliberately — produces a system that is at once simple to reason about and powerful in practice. Each tool earns its place: Snowflake separates storage from compute, dbt makes transformations testable, Airflow turns the pipeline into a deployable artefact, and Tableau collapses the gap between insight and decision.

The same blueprint scales from synthetic e-commerce data to real-world retail, fintech and SaaS workloads with minimal change.

Take-away

Faker → S3 → Snowpipe → Snowflake → dbt → Airflow → Tableau is more than a list of tools. It is a deliberate, decoupled, testable architecture that any data team can replicate.